

Liquidity and Volatility Spillovers in the Oil Market

A Time Series Analytics Study Using EViews and Python

Time Series Analytics Project | April 2026

Sample Period: April 1, 2016 – March 31, 2026

1. Objective

This study examines liquidity and volatility spillovers in the global crude oil market using daily data for West Texas Intermediate (WTI) and Brent crude oil futures over the period April 1, 2016 to March 31, 2026, covering ten complete Indian financial years and 2,457 trading observations after data cleaning.

The two benchmarks chosen — WTI and Brent — together account for the pricing of the majority of globally traded crude oil. Alongside price returns and realized volatility, trading volume for both WTI and Brent futures is used as a proxy for market liquidity. This enables a comprehensive analysis of not only volatility transmission but also liquidity spillovers between and within the two markets.

The study period is particularly significant as it encompasses three major global crises: the COVID-19 pandemic (March 2020), the Russia-Ukraine war (February 2022), and the Iran-related Middle East conflict (October 2023). These episodes provide a stress-testing environment for oil market dynamics and are explicitly controlled for through event dummy variables.

The specific objectives of this study are:

1. To test for stationarity in all return, volatility, and liquidity series using the Augmented Dickey-Fuller (ADF) unit root test.
2. To estimate a Vector Autoregression (VAR) model and identify the optimal lag structure using information criteria.
3. To examine Granger causality among price returns, realized volatility, and trading volume (liquidity) across WTI and Brent markets.
4. To quantify volatility and liquidity spillovers using Impulse Response Functions (IRF) and Forecast Error Variance Decomposition (FEVD), computing separate Diebold-Yilmaz (2012) Total Connectedness Indices (TCI) for returns, volatility, and liquidity to ensure methodological consistency across variable types.
5. To assess how geopolitical shocks and the COVID-19 pandemic have influenced oil market spillover dynamics over the sample period.

2. Literature Review

2.1 Diebold and Yilmaz (2012)

The foundational framework for measuring spillovers in this study is drawn from Diebold and Yilmaz (2012), "Better to Give than to Receive: Predictive Directional Measurement of Volatility Spillovers," published in the *International Journal of Forecasting* (28(1), 57–66). The authors introduce a spillover index derived from forecast error variance decompositions (FEVD) of a Vector Autoregression (VAR) model. Unlike earlier symmetric measures, this framework enables directional decomposition — identifying which markets transmit shocks and which receive them.

The total spillover index is calculated as the proportion of total forecast error variance attributable to cross-market shocks rather than own-market innovations. This methodology is particularly well-suited to the WTI-Brent system, where multiple benchmarks co-exist in a tightly integrated global market, and directly informs the empirical strategy of this study.

2.2 CESifo Working Paper 10989 (2024)

The second reference is CESifo Working Paper No. 10989 (2024), which investigates volatility spillovers between oil prices and the exchange rates of major oil-importing and oil-exporting nations using a DCC-GARCH-Connectedness approach on daily data spanning March 2018 to August 2023. The paper conducts sub-period analysis across four phases: the full sample, pre-COVID-19, during COVID-19, and during the Russia-Ukraine war period.

This paper provides direct empirical support for two key choices in our study. First, it demonstrates that spillover dynamics in oil markets vary significantly across crisis and non-crisis periods, justifying our inclusion of event dummy variables. Second, it establishes that oil price volatility transmits meaningfully across market participants and benchmarks, motivating our focus on cross-market spillovers between WTI and Brent.

3. Data and Methodology

3.1 Data Description

Daily price and volume data for WTI crude oil futures (ticker: CL=F) and Brent crude oil futures (ticker: BZ=F) were downloaded from Yahoo Finance using the yfinance Python library. The sample period runs from April 1, 2016 to March 31, 2026, aligning with Indian financial years. After removing missing observations and days with zero trading volume, the final dataset contains 2,457 daily observations.

WTI is the primary US benchmark traded on the New York Mercantile Exchange (NYMEX), while Brent is the international benchmark traded on ICE Futures Europe. Trading volume for both futures contracts is used as a proxy for market liquidity. WTI volume represents the number of futures contracts traded daily on NYMEX, providing a clean and standardized measure. Brent volume is sourced from ICE via Yahoo Finance, enabling direct cross-market liquidity comparison. The following variables are constructed for the empirical analysis:

Variable	Description	Market	Construction
WTI_Return	Daily log return	WTI	$\ln(\text{Price}_t / \text{Price}_{t-1})$
Brent_Return	Daily log return	Brent	$\ln(\text{Price}_t / \text{Price}_{t-1})$
WTI_Vol	Realized volatility	WTI	21-day rolling SD of WTI_Return $\times \sqrt{252}$
Brent_Vol	Realized volatility	Brent	21-day rolling SD of Brent_Return $\times \sqrt{252}$
WTI_Volume_Change	Liquidity proxy	WTI	$\ln(\text{Volume}_t / \text{Volume}_{t-1})$
Brent_Volume_Change	Liquidity proxy	Brent	$\ln(\text{Volume}_t / \text{Volume}_{t-1})$

Table 1: Variable Definitions

Realized volatility is constructed as the 21-day rolling standard deviation of daily log returns, annualized by multiplying by the square root of 252 trading days. The 21-day window corresponds to one trading month and is standard in the commodity volatility literature. Volume changes are log differences of daily trading volume, with zero-volume observations replaced by missing values prior to transformation.

3.2 Event Dummy Variables

Three binary dummy variables are included as exogenous regressors in the VAR to control for structural shocks arising from major geopolitical and pandemic events:

Dummy Variable	Period	Event
D_COVID	March 1, 2020 – June 30, 2021	COVID-19 Pandemic
D_RUSSIA	February 24, 2022 – September 30, 2023	Russia-Ukraine War
D_IRAN	October 7, 2023 onwards	Iran-Middle East Conflict

Table 2: Event Dummy Variables

3.3 Empirical Methodology

Step 1 — Stationarity Testing: The Augmented Dickey-Fuller (ADF) test is applied to each of the six endogenous variables with a constant and automatic lag selection based on the Schwarz Information Criterion (SIC), maximum 26 lags.

Step 2 — VAR Estimation and Lag Selection: A VAR model is estimated with six endogenous variables (WTI_Return, Brent_Return, WTI_Vol, Brent_Vol, WTI_Volume_Change, Brent_Volume_Change) and

three exogenous event dummies plus a constant. The optimal lag is selected by minimising the Schwarz Information Criterion tested up to 15 lags.

Step 3 — VAR Stability Check: Stability is verified by confirming all inverse roots of the AR characteristic polynomial lie within the unit circle.

Step 4 — Granger Causality: Block exogeneity Wald tests are conducted within the VAR framework to identify directional causality. As defined in the EViews User's Guide II (Chapter 44, p. 848): "For each equation in the VAR, the output displays χ^2 (Wald) statistics for the joint significance of each of the other lagged endogenous variables in that equation." The null hypothesis is that all lag coefficients of the excluded variable are jointly zero — meaning that variable adds no predictive power beyond what the rest of the system already explains. A variable X Granger-causes Y if lagged values of X significantly improve the one-step-ahead forecast of Y, after controlling for all other variables in the system.

Step 5 — Impulse Response Functions (IRF): Cholesky-decomposed impulse responses trace each variable's dynamic response to a one standard deviation shock over a 10-day horizon.

Step 6 — Variance Decomposition and Spillover Index: Following Diebold and Yilmaz (2012), FEVD at the 10-day horizon quantifies cross-market shock contributions. Three separate Total Connectedness Indices (TCI) are computed — one each for returns, volatility, and liquidity — by extracting the relevant variable pairs from the FEVD table. Each TCI equals the sum of off-diagonal FEVD shares for that variable type divided by the total FEVD for that pair, expressed as a percentage. This ensures that spillover indices are computed only between variables of the same economic nature.

4. Findings

4.1 Descriptive Statistics

Table 3 presents descriptive statistics for all variables over 2,457 daily observations. Mean log returns are approximately zero for both WTI (0.011%) and Brent (0.027%), consistent with the efficient market hypothesis for commodity futures. Both series exhibit significant negative skewness (−0.977 for WTI, −0.832 for Brent), indicating that large negative daily price movements occur more frequently than equivalent positive movements.

Kurtosis values of 20.74 (WTI) and 16.42 (Brent) far exceed the normal distribution benchmark of 3, confirming fat-tailed distributions. The Jarque-Bera statistic rejects normality at the 1% level for all

variables. The maximum single-day loss of -28.22% in WTI returns corresponds to the historic April 2020 price crash during COVID-19. Realized volatility peaked at 201.8% (WTI) and 168.1% (Brent) on an annualized basis during the pandemic. WTI trading volume averaged approximately 446,855 contracts per day, far exceeding Brent's average of 32,549, reflecting WTI's greater depth as a standardized NYMEX futures contract. Brent_Volume_Change exhibits higher proportional variability (std dev = 0.565) compared to WTI_Volume_Change (0.441), indicating greater liquidity volatility in the Brent market.

Statistic	WTI Ret.	Brent Ret.	WTI Vol	Brent Vol	WTI Vol.Ch.	Brent Vol.Ch.	N
Mean	0.0001	0.0003	0.3519	0.3263	-0.0012	-0.0004	2,457
Std Dev	0.0257	0.0235	0.1991	0.1647	0.4411	0.5647	
Minimum	-0.2822	-0.2758	0.1407	0.1167	-1.5213	-3.5358	
Maximum	0.2205	0.1908	2.0182	1.6806	2.1154	4.3829	
Skewness	-0.977	-0.832	4.351	3.083	1.558	1.683	
Kurtosis	20.74	16.42	29.66	18.13	8.474	19.08	
JB p-value	0.000***	0.000***	0.000***	0.000***	0.000***	0.000***	

Table 3: Descriptive Statistics (May 2016 – March 2026, $n = 2,457$)

*** denotes rejection of normality at 1% significance level (Jarque-Bera). Vol.Ch. = Volume Change.

4.2 Stationarity Tests

Table 4 reports ADF unit root test results for all six series. The null hypothesis of a unit root is rejected at the 1% significance level for all variables, confirming that the entire system is integrated of order zero, $I(0)$. This validates direct VAR estimation in levels without differencing and eliminates the need for cointegration analysis.

Return series require zero lags, indicating clean stationarity consistent with efficient market pricing. Volatility series require 22 lags, reflecting well-documented persistence in oil market volatility over monthly horizons. Volume change series require 23–24 lags, capturing weekly cyclical patterns in futures trading activity.

Series	ADF t-Statistic	p-value	Lags	Crit. Val. (1%)	Conclusion
WTI_Return	-51.839***	0.0001	0	-3.4328	I(0) ✓
Brent_Return	-49.249***	0.0001	0	-3.4328	I(0) ✓
WTI_Vol	-5.625***	0.0000	22	-3.4328	I(0) ✓
Brent_Vol	-5.041***	0.0000	22	-3.4328	I(0) ✓
WTI_Volume_Change	-13.144***	0.0000	24	-3.4328	I(0) ✓
Brent_Volume_Change	-15.086***	0.0000	23	-3.4328	I(0) ✓

Table 4: Augmented Dickey-Fuller Unit Root Test Results

*** denotes rejection of the unit root null hypothesis at the 1% significance level. Lag length selected automatically by SIC, maximum 26 lags. Exogenous: Constant.

4.3 VAR Lag Length Selection

The VAR model is estimated with six endogenous variables and information criteria computed for up to 15 lags. The Schwarz Information Criterion — which imposes a stricter penalty for additional parameters and is preferred for large samples — selects lag 4. With six endogenous variables, a lag of 15 would require estimating 540 VAR coefficients, introducing a substantial risk of overfitting. Lag 4 captures one full trading week and is adopted as the final specification.

Lag	LogL	LR	FPE	AIC	SC	HQ
1	21796.59	13644.69	7.48e-16	-17.802	-17.660	-17.750
4	22387.61	363.153	5.03e-16	-18.198	-17.799*	-18.053
12	23316.36	254.439	2.98e-16	-18.723	-17.640	-18.329*
15	23493.99	103.296*	2.81e-16*	-18.780*	-17.440	-18.293

Table 5: VAR Lag Order Selection Criteria (Selected Rows)

* indicates the lag selected by that criterion. Chosen lag: 4 (SC). Exogenous: C D_COVID D_RUSSIA D_IRAN.

4.4 VAR Stability and Model Fit

All inverse roots of the AR characteristic polynomial for the VAR(4) model have modulus strictly less than 1.0 and lie within the unit circle, confirming model stability and validating subsequent IRF and FEVD analysis.

The VAR(4) model yields markedly different explanatory power across equations. The volatility equations achieve R^2 values of 97.3% (Brent_Vol) and 97.8% (WTI_Vol), confirming strong own-volatility

persistence over the sample. In contrast, return equations show R^2 below 2.1% — Brent_Return at 1.4% and WTI_Return at 2.0% — consistent with the efficient market hypothesis where past information has little predictive power over future returns.

4.5 Granger Causality

Table 6 reports Granger causality block exogeneity Wald test results from the VAR(4) model. The test is conducted within the VAR framework as defined in EViews User's Guide II (Chapter 44, p. 848), which states that for each equation, the output displays Chi-square Wald statistics for the joint significance of each of the other lagged endogenous variables. The term 'Excluded' in the output refers to the one variable being temporarily removed from the full model — all other variables remain as controls. The null hypothesis is that all four lag coefficients of the excluded variable are jointly zero ($H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$). A Chi-square p-value below 0.05 leads to rejection of this null, confirming that the excluded variable Granger-causes the dependent variable even after controlling for the full system.

Dependent Variable	Excluded (Cause)	Chi-sq	p-value	Causality?
WTI_Return	Brent_Return	20.040	0.0005***	Yes ✓
WTI_Return	All others	40.868	0.0039***	System significant
Brent_Return	WTI_Vol	9.624	0.0473**	Yes ✓
WTI_Vol	WTI_Return	105.242	0.0000***	Yes ✓ (strong)
WTI_Vol	Brent_Return	94.450	0.0000***	Yes ✓ (strong)
WTI_Vol	Brent_Vol	150.283	0.0000***	Yes ✓ (strongest)
Brent_Vol	Brent_Return	14.363	0.0062***	Yes ✓
Brent_Vol	WTI_Return	12.297	0.0153**	Yes ✓
Brent_Vol	WTI_Vol	40.449	0.0000***	Yes ✓ (strong)
Brent_Vol	Brent_Vol_Change	14.938	0.0048***	Yes ✓ (Liquidity → Volatility)
Brent_Vol_Change	WTI_Vol_Change	13.296	0.0099***	Yes ✓ (Liquidity Spillover)
WTI_Vol_Change	All variables	24.104	0.2379	No X (system insignificant)

Table 6: VAR Granger Causality / Block Exogeneity Wald Tests

***, ** denote significance at the 1% and 5% levels respectively. $df = 4$ for all pairwise tests.

The results reveal several important transmission channels. First, Brent returns Granger-cause WTI returns ($p = 0.0005$) but the reverse is not significant, establishing a clear and unidirectional price

leadership role for Brent crude. Second, both WTI and Brent returns significantly Granger-cause WTI volatility ($p = 0.000$ for both), and both cause Brent volatility as well, confirming return-to-volatility spillovers in both directions. The largest chi-square statistic in the table (150.283) corresponds to the Brent_Vol $\rightarrow$ WTI_Vol channel, indicating that Brent volatility is the single strongest predictor of WTI volatility.

Third, Brent_Volume_Change significantly Granger-causes Brent_Vol ($p = 0.0048$), establishing a liquidity-to-volatility spillover channel within the Brent market. Fourth, WTI_Volume_Change Granger-causes Brent_Volume_Change ($p = 0.0099$), demonstrating a cross-market liquidity spillover. WTI_Volume_Change itself is not Granger-caused by any variable in the system, suggesting WTI liquidity is exogenously determined at this frequency.

4.6 Impulse Response Analysis

Table E4 presents the stacked impulse response functions from the VAR(4) model, tracing each variable's response to one standard deviation shocks from all variables over a 10-day horizon using Cholesky decomposition. The response of Brent_Return to all innovations remains negligible throughout the horizon, confirming its role as an exogenous price setter. WTI_Return shows a visible positive response to Brent_Return shocks, consistent with Brent's price leadership established in the Granger causality analysis.

The volatility panels reveal meaningful cross-market transmission. Brent_Vol responds positively to WTI_Vol shocks, and WTI_Vol absorbs shocks from multiple sources, peaking at period 1 and dissipating within 4 to 5 trading days, indicating transitory but significant volatility spillovers. Both volume change panels are dominated by their own shocks, with sharp initial spikes that decay rapidly, confirming that liquidity disturbances are short-lived and self-correcting. These patterns are fully consistent with the Granger causality findings and reinforce the directional nature of spillover transmission in the WTI-Brent system.

4.7 Variance Decomposition and Spillover Indices

Following Diebold and Yilmaz (2012), forecast error variance decompositions (FEVD) at the 10-day horizon are used to quantify the degree of cross-market shock transmission. The Total Connectedness Index (TCI) for each variable type is computed as:

$$TCI = \Sigma(\text{off-diagonal FEVD}) / \Sigma(\text{all FEVD}) \times 100$$

To ensure methodological consistency, separate TCIs are computed for each category of variable: returns, realized volatility, and trading volume changes. This approach ensures that spillover indices are computed only between economically comparable variables, making the results more interpretable and precise. The FEVD values used are drawn directly from the EViews output at the 10-day horizon (see Appendix Table F2).

4.7.1 Return Spillovers

Table 7 presents the FEVD-based spillover table for the return series. Brent_Return explains 86.74% of WTI_Return's forecast error variance, while WTI_Return explains only 0.10% of Brent_Return's variance. This extreme asymmetry confirms that Brent crude is the dominant price setter in the global oil market, transmitting price signals unidirectionally to WTI.

To \ From	Brent_Return	WTI_Return	From Others	To Others
Brent_Return	98.90	0.10	1.10%	86.84%
WTI_Return	86.74	12.55	87.45%	0.20%
TCI — Returns				43.42%

Table 7: Return Spillover Table (Period 10, Cholesky FEVD). All values in %.

$TCI_Returns = (0.10 + 86.74) / 200 \times 100 = 43.42\%$. Source: EViews FEVD output, Appendix Table F2.

4.7.2 Volatility Spillovers

Table 8 presents the volatility spillover table. This is the most important component of the analysis. Brent_Vol explains 59.86% of WTI_Vol's forecast error variance at the 10-day horizon, while WTI_Vol accounts for only 3.25% of Brent_Vol's variance. This strongly asymmetric result confirms that volatility transmission in the WTI-Brent system is predominantly unidirectional — from Brent to WTI — making Brent crude the dominant volatility transmitter in the global oil market.

To \ From	Brent_Vol	WTI_Vol	From Others	To Others
Brent_Vol	90.23	3.25	9.77%	63.11%
WTI_Vol	59.86	30.55	69.45%	3.25%
TCI — Volatility				31.56%

Table 8: Volatility Spillover Table (Period 10, Cholesky FEVD). All values in %.

$TCI_Volatility = (3.25 + 59.86) / 200 \times 100 = 31.56\%$. Highlighted cell (59.86%) is the key spillover finding. Source: EViews FEVD output, Appendix Table F2.

4.7.3 Liquidity Spillovers

Table 9 presents the liquidity spillover table using the volume change series as proxies for market liquidity. Both series are predominantly self-determined — Brent_Volume_Change at 98.15% and WTI_Volume_Change at 95.76% — leaving minimal cross-market contributions. The TCI of 1.37% confirms that liquidity spillovers, while statistically detectable through Granger causality in the short run, are quantitatively negligible over the 10-day forecast horizon. This finding is consistent with the block exogeneity results where volume change series showed no significant causality from price or volatility variables.

To \ From	Brent_VolCh	WTI_VolCh	From Others	To Others
Brent_VolCh	98.15	0.44	1.85%	2.73%
WTI_VolCh	2.29	95.76	4.24%	0.44%
TCI — Liquidity				1.37%

Table 9: Liquidity Spillover Table (Period 10, Cholesky FEVD). All values in %.

$TCI_Liquidity = (0.44 + 2.29) / 200 \times 100 = 1.37\%$. Source: EViews FEVD output, Appendix Table F2.

4.7.4 Summary of Spillover Indices

Table 10 summarises the three Total Connectedness Indices computed for each variable category. The results reveal a clear hierarchy of market integration: return spillovers are the strongest, followed by volatility spillovers, while liquidity spillovers are negligible. This pattern is consistent with financial market theory — prices integrate fastest, volatility transmits with a short lag, and trading activity is largely determined by each market's own microstructure.

Variable Type	TCI (%)	Dominant Direction	Key Finding
Returns	43.42%	Brent → WTI	Brent explains 86.74% of WTI return variance
Volatility	31.56%	Brent → WTI	Brent explains 59.86% of WTI volatility variance
Liquidity	1.37%	Minimal	Volume largely self-contained in each market

Table 10: Summary of Diebold-Yilmaz Total Connectedness Indices by Variable Type

All TCIs computed from EViews Cholesky FEVD at 10-day horizon. Diebold and Yilmaz (2012) framework applied separately for each variable category.

Across all three spillover dimensions, Brent crude consistently emerges as the dominant market. It leads WTI in price discovery (TCI_Returns = 43.42%), drives WTI volatility (TCI_Volatility = 31.56%), and is the most insulated from external liquidity shocks (Brent_VolCh = 98.15% own-variance). WTI, by contrast, is

a price-taker and volatility-receiver, with 87.45% of its return variance and 69.45% of its volatility variance attributable to external sources — primarily Brent. The negligible liquidity TCI of 1.37% confirms that while WTI volume Granger-causes Brent volume in the short run (as shown in Section 4.5), this relationship does not translate into meaningful forecast error variance transmission at the 10-day horizon.

4.8 Role of Geopolitical Events

The event dummy variables for COVID-19, Russia-Ukraine, and the Iran conflict are included as exogenous regressors but are individually insignificant in most VAR equations. This is an economically meaningful finding: the extraordinary price dislocations and volatility spikes associated with each crisis are absorbed endogenously within the VAR's volatility persistence dynamics rather than as discrete structural breaks.

Figure 1 provides clear visual corroboration. The COVID-19 crash of March–April 2020 produced the most extreme volatility episode in the sample, with WTI realized volatility reaching 201.8% annualized and daily returns falling as low as -28.22% . The Russia-Ukraine war generated a secondary volatility peak of approximately 75%, driven by supply-side uncertainty. The Iran conflict from October 2023 shows a moderate but sustained elevation in volatility visible into early 2026. All three episodes are captured within the time-varying dynamics of the VAR(4) system, demonstrating the model's robustness across crisis and non-crisis periods.

5. Conclusion

This study investigates liquidity and volatility spillovers within the global crude oil market using daily data for WTI and Brent crude oil futures across ten Indian financial years from April 2016 to March 2026. Employing a VAR-based framework anchored in the Diebold-Yilmaz (2012) spillover index methodology, the analysis covers stationarity testing, Granger causality, impulse response analysis, and forecast error variance decomposition for a six-variable system encompassing returns, realized volatility, and trading volume changes for both benchmarks.

Six conclusions emerge. First, all six series are stationary at level — $I(0)$ — confirming the appropriateness of VAR estimation in levels. Second, Brent crude serves as the unidirectional price leader: Brent returns Granger-cause WTI returns, but not vice versa. Third, significant and asymmetric volatility spillovers exist between the markets. Brent volatility explains 59.86% of WTI volatility's forecast error variance at the 10-day horizon, while WTI explains only 3.25% of Brent volatility — establishing

Brent as the dominant volatility transmitter. Separate Total Connectedness Indices computed by variable type reveal a clear hierarchy: return spillovers are strongest (TCI = 43.42%), followed by volatility spillovers (TCI = 31.56%), while liquidity spillovers are negligible (TCI = 1.37%).

Fourth, impulse responses confirm that volatility shocks are transitory, dissipating within four to five trading days. Fifth, Brent trading volume Granger-causes Brent volatility ($p = 0.0048$), establishing a statistically significant liquidity-to-volatility spillover channel, and WTI trading volume Granger-causes Brent trading volume ($p = 0.0099$), confirming cross-market liquidity transmission. These liquidity spillover findings, enabled by the inclusion of both markets' volume data, represent an important contribution of this study. Sixth, geopolitical shocks are endogenously absorbed by the VAR's persistence structure, with all crisis episodes — COVID-19, Russia-Ukraine, and the Iran conflict — reflected in elevated volatility dynamics rather than discrete structural breaks.

These findings carry practical implications for risk managers, portfolio hedgers, and energy market policymakers. The dominant role of Brent volatility in driving WTI volatility (TCI_Volatility = 31.56%) implies that monitoring Brent market conditions is essential for managing WTI exposure. The liquidity-to-volatility spillover from Brent volume identified through Granger causality suggests that sharp changes in Brent trading activity can serve as an early indicator of impending volatility shifts across both benchmarks. The strong return TCI of 43.42% confirms near-complete price integration between the two benchmarks. Future research could employ a rolling-window Diebold-Yilmaz framework to examine how each of the three TCIs evolves across the distinct crisis regimes identified in this study.

References

Diebold, F.X. and Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28(1), 57–66.

CESifo (2024). Volatility spillover between oil prices and main exchange rates: Evidence from a DCC-GARCH-connectedness approach. *CESifo Working Paper No. 10989*. Munich: CESifo Institute.

Yahoo Finance (2026). Historical data: WTI Crude Oil Futures (CL=F) and Brent Crude Oil Futures (BZ=F). Retrieved via yfinance Python library, April 2026.

IHS Global Inc. (2020). *EViews 11 User's Guide II*. Chapter 44: Vector Autoregression and Error Correction Models, p. 848. Irvine, CA: IHS Global Inc.

Appendix: Output Evidence Index

The following index provides labeled placeholders for attaching printed screenshots of all EViews and Python outputs referenced in the report body.

A. Python Outputs

Figure 1 WTI & Brent Prices, Log Returns and Realized Volatility — 3-panel time series plot with COVID-19, Russia-Ukraine and Iran event markers

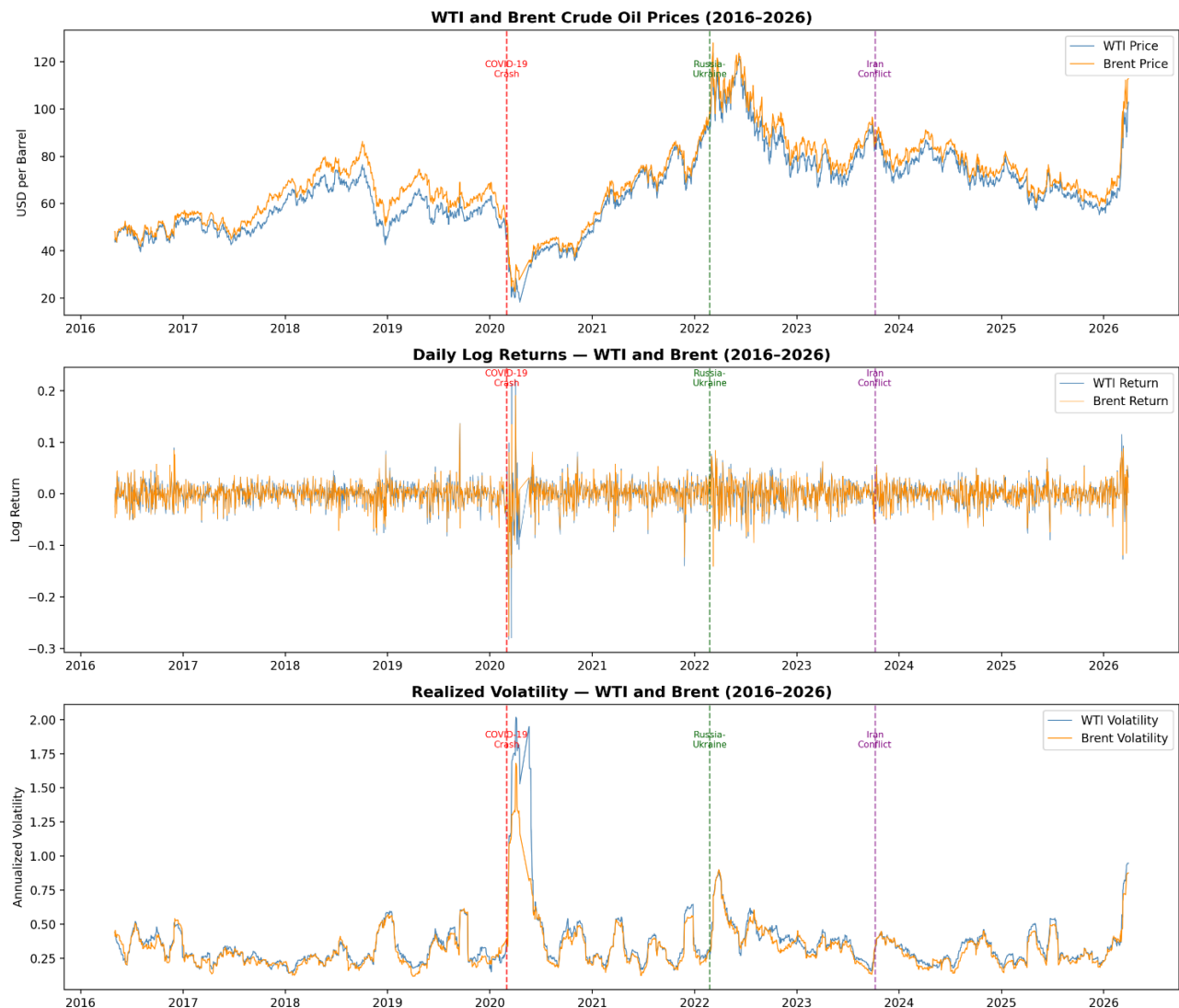


Figure 2 Python Data Output — Final dataset shape (2457 × 13), tail(), missing value check, and descriptive stats from Jupyter Notebook

```

--- Missing Values ---
WTI_Price      0
WTI_Volume     0
Brent_Price    0
Brent_Volume   0
WTI_Return     0
Brent_Return   0
WTI_Vol_Change 0
Brent_Vol_Change 0
WTI_Vol        0
Brent_Vol      0
D_COVID        0
D_RUSSIA       0
D_IRAN         0
dtype: int64

```

B. EViews — Descriptive Statistics

Table B1 Descriptive Statistics — WTI Series: WTI_Price, WTI_Return, WTI_Vol, WTI_Volume, WTI_Volume_Change

Date: 04/26/26 Time: 16:40 Sample: 5/04/2016 3/30/2026					
	WTI_RETURN	WTI_PRICE	WTI_VOL	WTI_VOLUME	WTI_VOLUME_CHANGE
Mean	0.000112	65.14094	0.351927	446855.3	-0.001234
Median	0.002114	64.38000	0.301070	397868.0	-0.031075
Maximum	0.220481	123.7000	2.018177	1774383.	2.115387
Minimum	-0.282206	18.27000	0.140673	58714.00	-1.521252
Std. Dev.	0.025676	16.46979	0.199073	218875.8	0.441078
Skewness	-0.977350	0.460521	4.350768	0.745175	1.557733
Kurtosis	20.73916	3.537150	29.65984	3.695694	8.473655
Jarque-Bera	32606.30	116.3847	80514.24	276.9380	4060.912
Probability	0.000000	0.000000	0.000000	0.000000	0.000000
Sum	0.275641	160051.3	864.6852	1.10E+09	-3.031047
Sum Sq. Dev.	1.619148	666200.1	97.33183	1.18E+14	477.8148
Observations	2457	2457	2457	2457	2457

Table B2 Descriptive Statistics — Brent Series: Brent_Price, Brent_Return, Brent_Vol, Brent_Volume, Brent_Volume_Change

Date: 04/26/26 Time: 16:40 Sample: 5/04/2016 3/30/2026					
	BRENT_PRICE	BRENT_RETURN	BRENT_VOL	BRENT_VOLUME	BRENT_VOLUME_CHANGE
Mean	69.51125	0.000272	0.326267	32548.57	-0.000444
Median	69.05000	0.001632	0.283282	30734.00	-0.034654
Maximum	127.9800	0.190774	1.680595	235965.0	4.382944
Minimum	22.74000	-0.275751	0.116744	387.0000	-3.535760
Std. Dev.	16.85815	0.023465	0.164676	16607.51	0.564655
Skewness	0.338168	-0.831887	3.082687	3.368009	1.683038
Kurtosis	3.335687	16.42485	18.12687	30.97322	19.07594
Jarque-Bera Probability	58.36563 0.000000	18734.10 0.000000	27317.13 0.000000	84753.69 0.000000	27617.33 0.000000
Sum	170789.1	0.668064	801.6372	79971833	-1.091161
Sum Sq. Dev.	697988.0	1.352260	66.60231	6.77E+11	783.0588
Observations	2457	2457	2457	2457	2457

C. EViews — ADF Unit Root Tests

Table C1 ADF Test: WTI_Return — t = -51.839, p = 0.0001, Lag = 0

Augmented Dickey-Fuller Unit Root Test on WTI_RETURN

Null Hypothesis: WTI_RETURN has a unit root				
Exogenous: Constant				
Lag Length: 0 (Automatic - based on SIC, maxlag=26)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-51.83904	0.0001
Test critical values:	1% level		-3.432818	
	5% level		-2.862516	
	10% level		-2.567335	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(WTI_RETURN)				
Method: Least Squares				
Date: 04/26/26 Time: 16:59				
Sample (adjusted): 5/05/2016 3/30/2026				
Included observations: 2456 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
WTI_RETURN(-1)	-1.045687	0.020172	-51.83904	0.0000
C	0.000116	0.000518	0.223173	0.8234
R-squared	0.522688	Mean dependent var		1.18E-05
Adjusted R-squared	0.522493	S.D. dependent var		0.037133
S.E. of regression	0.025660	Akaike info criterion		-4.486977
Sum squared resid	1.615762	Schwarz criterion		-4.482249
Log likelihood	5512.008	Hannan-Quinn criter.		-4.485259
F-statistic	2687.286	Durbin-Watson stat		1.998380
Prob(F-statistic)	0.000000			

Table C2 ADF Test: Brent_Return — t = -49.249, p = 0.0001, Lag = 0

Augmented Dickey-Fuller Unit Root Test on BRENT_RETURN

Null Hypothesis: BRENT RETURN has a unit root				
Exogenous: Constant				
Lag Length: 0 (Automatic - based on SIC, maxlag=26)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-49.24940	0.0001
Test critical values:	1% level		-3.432818	
	5% level		-2.862516	
	10% level		-2.567335	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(BRENT_RETURN)				
Method: Least Squares				
Date: 04/26/26 Time: 16:58				
Sample (adjusted): 5/05/2016 3/30/2026				
Included observations: 2456 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
BRENT_RETURN(-1)	-0.993347	0.020170	-49.24940	0.0000
C	0.000289	0.000473	0.611008	0.5413
R-squared	0.497080	Mean dependent var		1.97E-05
Adjusted R-squared	0.496875	S.D. dependent var		0.033067
S.E. of regression	0.023455	Akaike info criterion		-4.666679
Sum squared resid	1.349999	Schwarz criterion		-4.661951
Log likelihood	5732.682	Hannan-Quinn criter.		-4.664961
F-statistic	2425.504	Durbin-Watson stat		1.999468
Prob(F-statistic)	0.000000			

Table C3 ADF Test: WTI_Vol — t = -5.625, p = 0.0000, Lag = 22

Augmented Dickey-Fuller Unit Root Test on WTI_VOL

Null Hypothesis: WTI_VOL has a unit root				
Exogenous: Constant				
Lag Length: 22 (Automatic - based on SIC, maxlag=26)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-5.625084	0.0000
Test critical values:	1% level		-3.432842	
	5% level		-2.862527	
	10% level		-2.567341	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(WTI_VOL)				
Method: Least Squares				
Date: 04/26/26 Time: 17:00				
Sample (adjusted): 6/07/2016 3/30/2026				
Included observations: 2434 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
WTI_VOL(-1)	-0.020694	0.003679	-5.625084	0.0000
D(WTI_VOL(-1))	0.048657	0.020267	2.400809	0.0164
D(WTI_VOL(-2))	0.067575	0.019455	3.473412	0.0005
D(WTI_VOL(-3))	0.114477	0.019461	5.882230	0.0000
D(WTI_VOL(-4))	-0.058919	0.019591	-3.007518	0.0027
D(WTI_VOL(-5))	0.093906	0.019591	4.793279	0.0000
D(WTI_VOL(-6))	0.060879	0.019688	3.092230	0.0020
D(WTI_VOL(-7))	0.072328	0.019724	3.667088	0.0003
D(WTI_VOL(-8))	0.039266	0.019781	1.984976	0.0473
D(WTI_VOL(-9))	0.084151	0.019780	4.254420	0.0000
D(WTI_VOL(-10))	-0.012901	0.019848	-0.649971	0.5158
D(WTI_VOL(-11))	0.056688	0.019841	2.857136	0.0043
D(WTI_VOL(-12))	0.051452	0.019873	2.589017	0.0097
D(WTI_VOL(-13))	-0.009126	0.019892	-0.458777	0.6464
D(WTI_VOL(-14))	-0.026087	0.019862	-1.313468	0.1892
D(WTI_VOL(-15))	0.034293	0.019991	1.715406	0.0864
D(WTI_VOL(-16))	0.005534	0.019984	0.276906	0.7819
D(WTI_VOL(-17))	-0.027666	0.020018	-1.382012	0.1671
D(WTI_VOL(-18))	0.078957	0.019973	3.953220	0.0001
D(WTI_VOL(-19))	0.049737	0.019979	2.489437	0.0129
D(WTI_VOL(-20))	-0.039820	0.019915	-1.999487	0.0457
D(WTI_VOL(-21))	-0.265528	0.019919	-13.33063	0.0000
D(WTI_VOL(-22))	0.064387	0.020629	3.121102	0.0018
C	0.007421	0.001421	5.220824	0.0000
R-squared	0.149406	Mean dependent var		0.000288
Adjusted R-squared	0.141289	S.D. dependent var		0.031825
S.E. of regression	0.029491	Akaike info criterion		-4.199649
Sum squared resid	2.096029	Schwarz criterion		-4.142486
Log likelihood	5134.973	Hannan-Quinn criter.		-4.178868
F-statistic	18.40500	Durbin-Watson stat		2.001138
Prob(F-statistic)	0.000000			

Table C4 ADF Test: Brent_Vol — t = -5.041, p = 0.0000, Lag = 22

Augmented Dickey-Fuller Unit Root Test on BRENT_VOL

Null Hypothesis: BRENT_VOL has a unit root				
Exogenous: Constant				
Lag Length: 22 (Automatic - based on SIC, maxlag=26)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-5.040526	0.0000
Test critical values:	1% level		-3.432842	
	5% level		-2.862527	
	10% level		-2.567341	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(BRENT_VOL)				
Method: Least Squares				
Date: 04/26/26 Time: 16:59				
Sample (adjusted): 6/07/2016 3/30/2026				
Included observations: 2434 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
BRENT_VOL(-1)	-0.018776	0.003725	-5.040526	0.0000
D(BRENT_VOL(-1))	0.143704	0.020289	7.082919	0.0000
D(BRENT_VOL(-2))	0.030489	0.018574	1.641518	0.1008
D(BRENT_VOL(-3))	0.033890	0.018583	1.823690	0.0683
D(BRENT_VOL(-4))	0.010288	0.018588	0.553488	0.5800
D(BRENT_VOL(-5))	0.060452	0.018567	3.255905	0.0011
D(BRENT_VOL(-6))	0.015351	0.018698	0.820985	0.4117
D(BRENT_VOL(-7))	0.057314	0.018708	3.063615	0.0022
D(BRENT_VOL(-8))	0.097618	0.018743	5.208369	0.0000
D(BRENT_VOL(-9))	0.030426	0.018847	1.614337	0.1066
D(BRENT_VOL(-10))	0.002146	0.018859	0.113806	0.9094
D(BRENT_VOL(-11))	-0.023414	0.018855	-1.241812	0.2144
D(BRENT_VOL(-12))	0.031386	0.018832	1.666638	0.0957
D(BRENT_VOL(-13))	0.021157	0.018845	1.122716	0.2617
D(BRENT_VOL(-14))	0.033479	0.018856	1.775517	0.0759
D(BRENT_VOL(-15))	0.003327	0.018994	0.175140	0.8610
D(BRENT_VOL(-16))	0.010920	0.018987	0.575104	0.5653
D(BRENT_VOL(-17))	0.031026	0.019009	1.632148	0.1028
D(BRENT_VOL(-18))	0.067328	0.019003	3.543008	0.0004
D(BRENT_VOL(-19))	0.045798	0.019091	2.398920	0.0165
D(BRENT_VOL(-20))	0.014415	0.019110	0.754299	0.4507
D(BRENT_VOL(-21))	-0.413172	0.019120	-21.60922	0.0000
D(BRENT_VOL(-22))	0.087346	0.020714	4.216864	0.0000
C	0.006247	0.001310	4.770195	0.0000
R-squared	0.220633	Mean dependent var		0.000241
Adjusted R-squared	0.213195	S.D. dependent var		0.028044
S.E. of regression	0.024876	Akaike info criterion		-4.540052
Sum squared resid	1.491289	Schwarz criterion		-4.482889
Log likelihood	5549.244	Hannan-Quinn criter.		-4.519271
F-statistic	29.66314	Durbin-Watson stat		1.998997
Prob(F-statistic)	0.000000			

Table C5 ADF Test: WTI_Volume_Change — t = -13.144, p = 0.0000, Lag = 24

Augmented Dickey-Fuller Unit Root Test on WTI_VOLUME_CHANGE

Null Hypothesis: WTI_VOLUME_CHANGE has a unit root				
Exogenous: Constant				
Lag Length: 24 (Automatic - based on SIC, maxlag=26)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-13.14420	0.0000
Test critical values:	1% level		-3.432844	
	5% level		-2.862528	
	10% level		-2.567341	
*Mackinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(WTI_VOLUME_CHANGE)				
Method: Least Squares				
Date: 04/26/26 Time: 16:56				
Sample (adjusted): 6/09/2016 3/30/2026				
Included observations: 2432 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
WTI_VOLUME_CHANGE(-1)	-6.318284	0.480690	-13.14420	0.0000
D(WTI_VOLUME_CHANGE(-1))	4.855349	0.475266	10.21607	0.0000
D(WTI_VOLUME_CHANGE(-2))	4.226579	0.466868	9.053052	0.0000
D(WTI_VOLUME_CHANGE(-3))	3.738071	0.454546	8.223743	0.0000
D(WTI_VOLUME_CHANGE(-4))	3.334408	0.438538	7.603462	0.0000
D(WTI_VOLUME_CHANGE(-5))	2.999099	0.419466	7.149802	0.0000
D(WTI_VOLUME_CHANGE(-6))	2.713762	0.398302	6.813329	0.0000
D(WTI_VOLUME_CHANGE(-7))	2.405863	0.376607	6.388261	0.0000
D(WTI_VOLUME_CHANGE(-8))	2.117734	0.355096	5.963832	0.0000
D(WTI_VOLUME_CHANGE(-9))	1.828310	0.333638	5.479925	0.0000
D(WTI_VOLUME_CHANGE(-10))	1.560362	0.312093	4.999665	0.0000
D(WTI_VOLUME_CHANGE(-11))	1.279980	0.290332	4.408680	0.0000
D(WTI_VOLUME_CHANGE(-12))	1.008089	0.268378	3.756225	0.0002
D(WTI_VOLUME_CHANGE(-13))	0.741420	0.246334	3.009815	0.0026
D(WTI_VOLUME_CHANGE(-14))	0.492202	0.224295	2.194445	0.0283
D(WTI_VOLUME_CHANGE(-15))	0.249380	0.202351	1.232411	0.2179
D(WTI_VOLUME_CHANGE(-16))	-0.012111	0.180397	-0.067135	0.9465
D(WTI_VOLUME_CHANGE(-17))	-0.291431	0.158609	-1.837414	0.0663
D(WTI_VOLUME_CHANGE(-18))	-0.597735	0.137051	-4.361402	0.0000
D(WTI_VOLUME_CHANGE(-19))	-0.799543	0.116161	-6.883072	0.0000
D(WTI_VOLUME_CHANGE(-20))	-0.790244	0.095642	-8.262485	0.0000
D(WTI_VOLUME_CHANGE(-21))	-0.638315	0.075353	-8.471054	0.0000
D(WTI_VOLUME_CHANGE(-22))	-0.404879	0.055310	-7.320208	0.0000
D(WTI_VOLUME_CHANGE(-23))	-0.209806	0.035974	-5.832210	0.0000
D(WTI_VOLUME_CHANGE(-24))	-0.066347	0.020359	-3.258866	0.0011
C	-0.006692	0.007204	-0.928941	0.3530
R-squared	0.710236	Mean dependent var	-7.95E-05	
Adjusted R-squared	0.707225	S.D. dependent var	0.654963	
S.E. of regression	0.354392	Akaike info criterion	0.773805	
Sum squared resid	302.1778	Schwarz criterion	0.835774	
Log likelihood	-914.9469	Hannan-Quinn criter.	0.796334	
F-statistic	235.8927	Durbin-Watson stat	2.005660	
Prob(F-statistic)	0.000000			

Table C6 ADF Test: Brent_Volume_Change — t = -15.086, p = 0.0000, Lag = 23

Augmented Dickey-Fuller Unit Root Test on BRENT_VOLUME_CHANGE

Null Hypothesis: BRENT_VOLUME_CHANGE has a unit root				
Exogenous: Constant				
Lag Length: 23 (Automatic - based on SIC, maxlag=26)				
			t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic			-15.08623	0.0000
Test critical values:	1% level		-3.432843	
	5% level		-2.862527	
	10% level		-2.567341	
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(BRENT_VOLUME_CHANGE)				
Method: Least Squares				
Date: 04/26/26 Time: 16:55				
Sample (adjusted): 6/08/2016 3/30/2026				
Included observations: 2433 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
BRENT_VOLUME_CHANGE(-1)	-7.789015	0.516300	-15.08623	0.0000
D(BRENT_VOLUME_CHANGE(-1))	6.158645	0.509885	12.07849	0.0000
D(BRENT_VOLUME_CHANGE(-2))	5.608673	0.498919	11.24164	0.0000
D(BRENT_VOLUME_CHANGE(-3))	5.072119	0.484278	10.47357	0.0000
D(BRENT_VOLUME_CHANGE(-4))	4.589939	0.466060	9.848382	0.0000
D(BRENT_VOLUME_CHANGE(-5))	4.144633	0.445124	9.311188	0.0000
D(BRENT_VOLUME_CHANGE(-6))	3.728233	0.422667	8.820734	0.0000
D(BRENT_VOLUME_CHANGE(-7))	3.330992	0.399721	8.333301	0.0000
D(BRENT_VOLUME_CHANGE(-8))	2.960951	0.376203	7.870618	0.0000
D(BRENT_VOLUME_CHANGE(-9))	2.617438	0.352300	7.429576	0.0000
D(BRENT_VOLUME_CHANGE(-10))	2.279354	0.327995	6.949364	0.0000
D(BRENT_VOLUME_CHANGE(-11))	1.943640	0.303422	6.405730	0.0000
D(BRENT_VOLUME_CHANGE(-12))	1.625166	0.278587	5.833603	0.0000
D(BRENT_VOLUME_CHANGE(-13))	1.315147	0.253570	5.186532	0.0000
D(BRENT_VOLUME_CHANGE(-14))	1.007320	0.228496	4.408479	0.0000
D(BRENT_VOLUME_CHANGE(-15))	0.715959	0.203345	3.520900	0.0004
D(BRENT_VOLUME_CHANGE(-16))	0.414390	0.178110	2.326599	0.0201
D(BRENT_VOLUME_CHANGE(-17))	0.125410	0.152963	0.819868	0.4124
D(BRENT_VOLUME_CHANGE(-18))	-0.182359	0.128190	-1.422566	0.1550
D(BRENT_VOLUME_CHANGE(-19))	-0.366812	0.104068	-3.524750	0.0004
D(BRENT_VOLUME_CHANGE(-20))	-0.402419	0.080949	-4.971286	0.0000
D(BRENT_VOLUME_CHANGE(-21))	-0.330851	0.059074	-5.600587	0.0000
D(BRENT_VOLUME_CHANGE(-22))	-0.239083	0.038932	-6.141102	0.0000
D(BRENT_VOLUME_CHANGE(-23))	-0.077929	0.020395	-3.821031	0.0001
C	0.001236	0.009408	0.131403	0.8955
R-squared	0.748755	Mean dependent var	-5.69E-05	
Adjusted R-squared	0.746251	S.D. dependent var	0.921153	
S.E. of regression	0.464017	Akaike info criterion	1.312431	
Sum squared resid	518.4705	Schwarz criterion	1.371996	
Log likelihood	-1571.572	Hannan-Quinn criter.	1.334085	
F-statistic	299.0117	Durbin-Watson stat	2.004024	
Prob(F-statistic)	0.000000			

D. EViews — VAR Model

Table D1 VAR Lag Order Selection Criteria — Up to 15 lags, SC selects Lag 4

VAR Lag Order Selection Criteria Endogenous variables: BRENT RETURN BRENT VOL BRENT VOLUME CHANG Exogenous variables: C D_COVID D_IRAN D_RUSSIA Date: 04/26/26 Time: 17:04 Sample: 5/04/2016 3/30/2026 Included observations: 2442						
Lag	LogL	LR	FPE	AIC	SC	HQ
0	14946.19	NA	1.98e-13	-12.22129	-12.16428	-12.20057
1	21796.59	13644.69	7.48e-16	-17.80228	-17.65976	-17.75048
2	22087.75	578.5022	6.07e-16	-18.01126	-17.78323	-17.92837
3	22203.92	230.2554	5.68e-16	-18.07692	-17.76338	-17.96296
4	22387.61	363.1530	5.03e-16	-18.19787	-17.79882*	-18.05283
5	22481.26	184.6929	4.80e-16	-18.24509	-17.76052	-18.06896
6	22562.08	158.9940	4.63e-16	-18.28180	-17.71172	-18.07459
7	22657.09	186.4529	4.41e-16	-18.33013	-17.67454	-18.09184
8	22841.56	361.0734	3.91e-16	-18.45173	-17.71062	-18.18235
9	22971.93	254.5530	3.62e-16	-18.52902	-17.70240	-18.22856
10	23048.22	148.5740	3.50e-16	-18.56201	-17.64988	-18.23047
11	23185.05	265.8226	3.22e-16	-18.64459	-17.64695	-18.28197
12	23316.36	254.4394	2.98e-16	-18.72265	-17.63950	-18.32895*
13	23381.23	125.3866	2.91e-16	-18.74630	-17.57763	-18.32151
14	23440.28	113.8399	2.86e-16	-18.76517	-17.51099	-18.30930
15	23493.99	103.2961*	2.81e-16*	-18.77968*	-17.43999	-18.29273
* indicates lag order selected by the criterion LR: sequential modified LR test statistic (each test at 5% level) FPE: Final prediction error AIC: Akaike information criterion SC: Schwarz information criterion HQ: Hannan-Quinn information criterion						

Table D2 VAR(4) Coefficient Estimates — Full equation output, R²: Brent_Vol = 97.3%, WTI_Vol = 97.8%

Vector Autoregression Estimates

Date: 04/28/26 Time: 17:47

Sample (adjusted): 5/10/2016 3/30/2026

Included observations: 2453 after adjustments

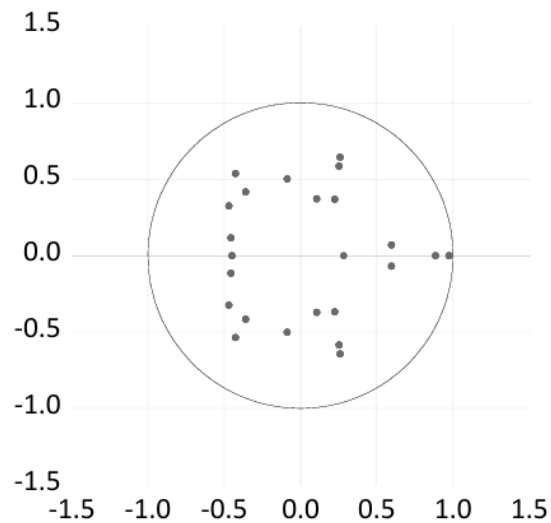
Standard errors in () & t-statistics in []

	BRENT RET	BRENT VOL	WTI RETURN	WTI VOL	BRENT VOL	WTI VOLUM
BRENT RETURN(-1)	0.025961 (0.05913) [0.43904]	0.038190 (0.06886) [0.55464]	0.259415 (0.06458) [4.01714]	0.554712 (0.07525) [7.37204]	-0.712872 (1.28885) [-0.55311]	-1.001679 (1.03035) [-0.97218]
BRENT RETURN(-2)	-0.014579 (0.06193) [-0.23541]	-0.086426 (0.07212) [-1.19839]	0.099839 (0.06764) [1.47610]	0.132804 (0.07881) [1.68510]	-2.776681 (1.34992) [-2.05692]	1.164839 (1.07917) [1.07939]
BRENT RETURN(-3)	-0.081874 (0.06210) [-1.31848]	-0.179131 (0.07231) [-2.47725]	-0.073101 (0.06782) [-1.07792]	-0.410893 (0.07902) [-5.19983]	-1.673543 (1.35351) [-1.23645]	0.209650 (1.08204) [0.19375]
BRENT_RETURN(-4)	0.068553 (0.05887) [1.16453]	-0.212620 (0.06855) [-3.10174]	0.039663 (0.06429) [0.61695]	-0.190802 (0.07491) [-2.54709]	0.685156 (1.28310) [0.53398]	0.220093 (1.02575) [0.21457]
BRENT VOL(-1)	0.017100 (0.02676) [0.63893]	1.001191 (0.03117) [32.1251]	0.010430 (0.02923) [0.35684]	0.238720 (0.03406) [7.00935]	-1.247741 (0.58336) [-2.13890]	-0.421096 (0.46635) [-0.90295]
BRENT VOL(-2)	0.035244 (0.03528) [0.99887]	-0.089387 (0.04109) [-2.17559]	0.036152 (0.03853) [0.93820]	-0.318115 (0.04490) [-7.08513]	0.505623 (0.76906) [0.65746]	0.177938 (0.61481) [0.28942]
BRENT VOL(-3)	-0.062458 (0.03507) [-1.78103]	0.001828 (0.04084) [0.04476]	-0.076922 (0.03830) [-2.00848]	-0.130037 (0.04463) [-2.91396]	0.476327 (0.76438) [0.62316]	-0.137772 (0.61107) [-0.22546]
BRENT_VOL(-4)	-0.000567 (0.02652) [-0.02139]	0.054668 (0.03088) [1.77037]	0.017362 (0.02896) [0.59951]	0.300766 (0.03374) [8.91296]	0.078505 (0.57800) [0.13582]	0.404017 (0.46207) [0.87436]
WTI_RETURN(-1)	-0.016211 (0.05411) [-0.29960]	-0.029807 (0.06301) [-0.47307]	-0.266996 (0.05909) [-4.51830]	-0.580801 (0.06885) [-8.43521]	0.668343 (1.17938) [0.56669]	1.163386 (0.94283) [1.23392]
WTI RETURN(-2)	0.028615 (0.05767) [0.49623]	0.029888 (0.06715) [0.44510]	-0.088775 (0.06298) [-1.40965]	-0.205619 (0.07338) [-2.80208]	2.092493 (1.25691) [1.66479]	-1.521736 (1.00481) [-1.51444]
WTI RETURN(-3)	0.029767 (0.05788) [0.51432]	0.125417 (0.06740) [1.86088]	0.009164 (0.06321) [0.14498]	0.318521 (0.07365) [4.32474]	1.968148 (1.26154) [1.56012]	-0.147552 (1.00851) [-0.14631]
WTI RETURN(-4)	-0.051653 (0.05486) [-0.94161]	0.209818 (0.06388) [3.28464]	-0.019003 (0.05991) [-0.31720]	0.151911 (0.06981) [2.17618]	-0.607115 (1.19569) [-0.50775]	-0.210103 (0.95587) [-0.21980]
WTI_VOL(-1)	0.001758 (0.02383) [0.07379]	0.139729 (0.02775) [5.03610]	0.008514 (0.02602) [0.32719]	0.861497 (0.03032) [28.4133]	0.970201 (0.51934) [1.86813]	-0.350122 (0.41518) [-0.84330]
WTI VOL(-2)	-0.054538 (0.02994) [-1.82166]	-0.043237 (0.03486) [-1.24022]	-0.073260 (0.03270) [-2.24062]	0.241518 (0.03810) [6.33943]	-0.902392 (0.65256) [-1.38285]	0.653629 (0.52168) [1.25294]

WTI_VOL(-3)	0.016050 (0.02987) [0.53742]	0.002275 (0.03478) [0.06542]	0.060343 (0.03262) [1.85012]	0.128438 (0.03800) [3.37959]	-0.312652 (0.65096) [-0.48029]	-0.406565 (0.52040) [-0.78126]
WTI_VOL(-4)	0.046966 (0.02340) [2.00733]	-0.085589 (0.02725) [-3.14143]	0.008863 (0.02555) [0.34685]	-0.323368 (0.02977) [-10.8610]	0.328002 (0.50998) [0.64317]	-0.013057 (0.40769) [-0.03203]
BRENT_VOLUME_CHA	-0.001034 (0.00094) [-1.10488]	0.003036 (0.00109) [2.78652]	0.000498 (0.00102) [0.48760]	0.001400 (0.00119) [1.17588]	-0.450279 (0.02039) [-22.0801]	-0.026183 (0.01630) [-1.60604]
BRENT_VOLUME_CHA	0.002173 (0.00101) [2.16184]	-0.001526 (0.00117) [-1.30361]	0.002395 (0.00110) [2.18192]	-0.000108 (0.00128) [-0.08441]	-0.289567 (0.02191) [-13.2162]	0.001165 (0.01752) [0.06651]
BRENT_VOLUME_CHA	0.000556 (0.00101) [0.55281]	-8.78E-05 (0.00117) [-0.07491]	0.000442 (0.00110) [0.40254]	0.000903 (0.00128) [0.70528]	-0.229607 (0.02194) [-10.4665]	-0.005659 (0.01754) [-0.32270]
BRENT_VOLUME_CHA	-8.15E-05 (0.00094) [-0.08657]	-0.000539 (0.00110) [-0.49223]	-0.000365 (0.00103) [-0.35541]	-0.000395 (0.00120) [-0.33009]	-0.131631 (0.02051) [-6.41801]	0.008478 (0.01640) [0.51709]
WTI_VOLUME_CHANG	-0.001062 (0.00117) [-0.90470]	-0.000783 (0.00137) [-0.57269]	-0.000971 (0.00128) [-0.75716]	-0.001096 (0.00149) [-0.73396]	-0.013882 (0.02559) [-0.54258]	-0.201495 (0.02045) [-9.85139]
WTI_VOLUME_CHANG	-0.000166 (0.00118) [-0.14128]	0.001474 (0.00137) [1.07594]	-0.000387 (0.00129) [-0.30078]	-1.66E-05 (0.00150) [-0.01106]	-0.016866 (0.02565) [-0.65750]	-0.380136 (0.02051) [-18.5371]
WTI_VOLUME_CHANG	-2.91E-05 (0.00118) [-0.02479]	-0.000332 (0.00137) [-0.24275]	0.000179 (0.00128) [0.13933]	-0.001281 (0.00150) [-0.85673]	-0.072917 (0.02562) [-2.84629]	-0.195463 (0.02048) [-9.54403]
WTI_VOLUME_CHANG	0.000475 (0.00117) [0.40564]	0.001565 (0.00136) [1.14864]	0.000140 (0.00128) [0.10943]	0.001228 (0.00149) [0.82495]	-0.066385 (0.02550) [-2.60340]	-0.146568 (0.02039) [-7.18994]
C	0.000223 (0.00120) [0.18645]	0.005436 (0.00139) [3.90251]	0.002522 (0.00131) [1.93060]	0.001751 (0.00152) [1.15016]	0.031816 (0.02607) [1.22032]	0.029667 (0.02084) [1.42341]
D_COVID	-0.000154 (0.00159) [-0.09724]	0.001319 (0.00185) [0.71326]	0.000282 (0.00173) [0.16280]	0.004177 (0.00202) [2.06703]	0.010877 (0.03461) [0.31427]	0.007715 (0.02767) [0.27885]
D_IRAN	0.000168 (0.00118) [0.14253]	0.000507 (0.00137) [0.36965]	0.000191 (0.00129) [0.14855]	0.001386 (0.00150) [0.92503]	-0.010376 (0.02567) [-0.40419]	-0.005315 (0.02052) [-0.25899]
D_RUSSIA	-0.000325 (0.00140) [-0.23168]	0.001226 (0.00163) [0.75022]	0.000689 (0.00153) [0.44977]	0.000636 (0.00179) [0.35637]	0.007676 (0.03058) [0.25100]	0.009228 (0.02445) [0.37742]
R-squared	0.014111	0.972918	0.020177	0.977880	0.193232	0.155204
Adj. R-squared	0.003134	0.972616	0.009268	0.977634	0.184250	0.145798
Sum sq. resid	1.329350	1.802577	1.585512	2.152651	631.5666	403.6272
S.E. equation	0.023413	0.027264	0.025570	0.029794	0.510333	0.407976
F-statistic	1.285512	3226.584	1.849553	3970.614	21.51189	16.50058
Log likelihood	5743.086	5369.584	5526.956	5151.901	-1816.463	-1267.345
Akaike AIC	-4.659670	-4.355144	-4.483454	-4.177661	1.503843	1.056131
Schwarz SC	-4.593408	-4.288881	-4.417191	-4.111398	1.570105	1.122393
Mean dependent	0.000301	0.326102	0.000114	0.351844	-0.000620	-0.001319
S.D. dependent	0.023450	0.164758	0.025689	0.199223	0.565035	0.441422
Determinant resid covariance (dof adj.)		4.68E-16				
Determinant resid covariance		4.37E-16				
Log likelihood		22494.24				
Akaike information criterion		-18.20322				
Schwarz criterion		-17.80564				
Number of coefficients		168				

Figure D3 AR Characteristic Polynomial Roots Graph — All roots inside unit circle, VAR(4) stable

Inverse Roots of AR Characteristic Polynomial



E. EViews — Granger Causality & Impulse Response Function

Table E1 Granger Causality — Dependent: WTI_Return and Brent_Return

Dependent variable: WTI RETURN			
Excluded	Chi-sq	df	Prob.
BRENT_RETURN	20.04000	4	0.0005
BRENT_VOL	5.653247	4	0.2266
BRENT_VOLUME_CH	5.725413	4	0.2206
WTI_VOL	7.676596	4	0.1042
WTI_VOLUME_CHAN	0.864300	4	0.9296
All	40.86805	20	0.0039
Dependent variable: BRENT RETURN			
Excluded	Chi-sq	df	Prob.
BRENT_VOL	6.137210	4	0.1891
BRENT_VOLUME_CH	9.348714	4	0.0530
WTI_RETURN	1.882759	4	0.7573
WTI_VOL	9.624231	4	0.0473
WTI_VOLUME_CHAN	1.214189	4	0.8758
All	29.58473	20	0.0769

Table E2 Granger Causality — Dependent: WTI_Vol and Brent_Vol

Dependent variable: WTI_VOL			
Excluded	Chi-sq	df	Prob.
BRENT_RETURN	94.44959	4	0.0000
BRENT_VOL	150.2826	4	0.0000
WTI_RETURN	105.2420	4	0.0000
BRENT_VOLUME_C...	2.716319	4	0.6064
WTI_VOLUME_CHA...	2.084819	4	0.7202
All	258.3327	20	0.0000

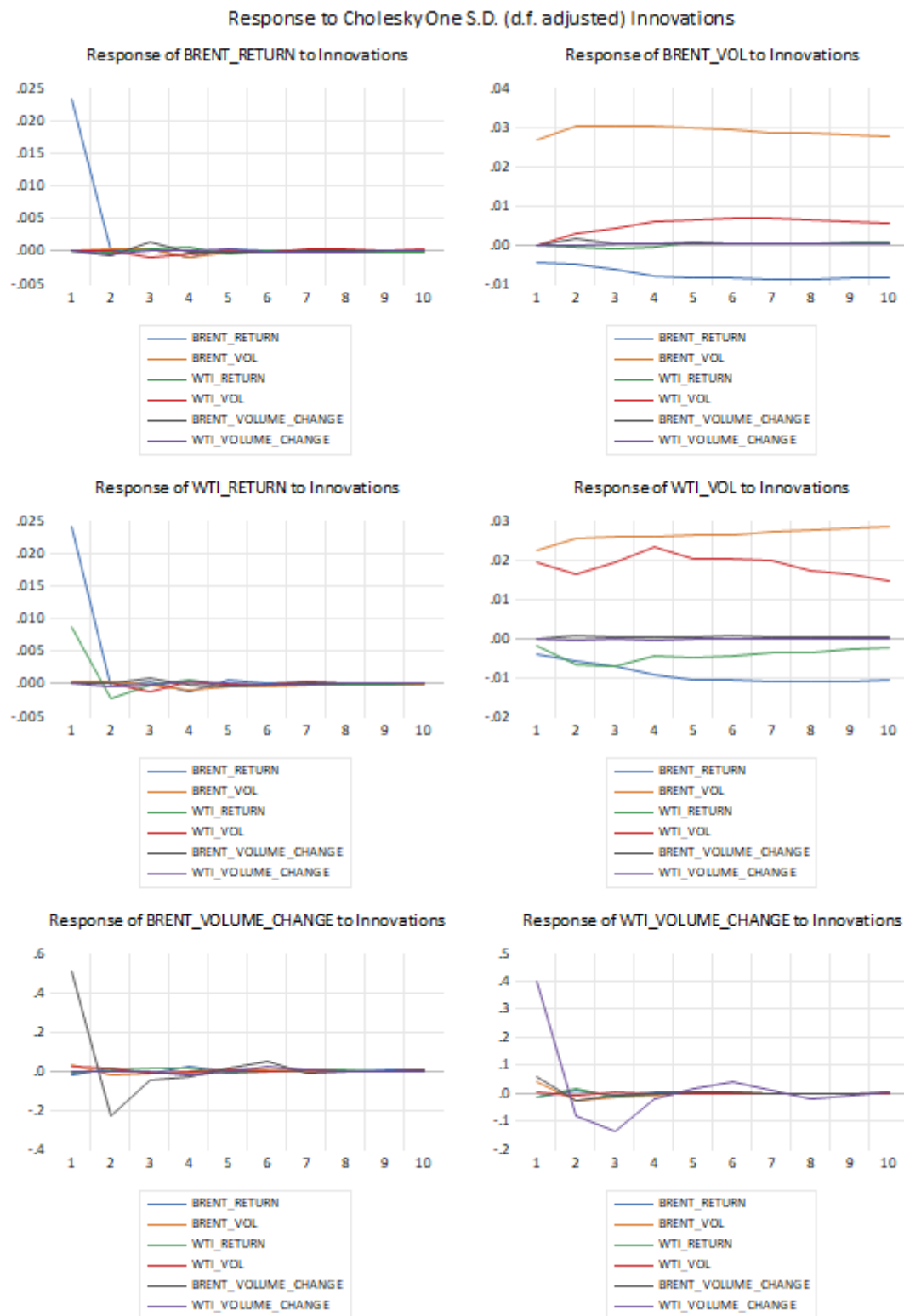
Dependent variable: BRENT_VOL			
Excluded	Chi-sq	df	Prob.
BRENT_RETURN	14.36263	4	0.0062
BRENT_VOLUME_CH	14.93754	4	0.0048
WTI_RETURN	12.29691	4	0.0153
WTI_VOL	40.44949	4	0.0000
WTI_VOLUME_CHAN	2.719410	4	0.6058
All	82.43537	20	0.0000

Table E3 Granger Causality — Dependent: WTI_Volume_Change and Brent_Volume_Change

Dependent variable: WTI VOLUME CHANGE			
Excluded	Chi-sq	df	Prob.
BRENT_RETURN	2.665011	4	0.6154
BRENT VOL	1.766977	4	0.7785
BRENT VOLUME CH	3.830239	4	0.4295
WTI_RETURN	5.198229	4	0.2676
WTI VOL	2.248409	4	0.6902
All	24.10453	20	0.2379

Dependent variable: BRENT VOLUME CHANGE			
Excluded	Chi-sq	df	Prob.
BRENT_RETURN	5.603611	4	0.2308
BRENT VOL	5.515846	4	0.2383
WTI_RETURN	5.004541	4	0.2868
WTI_VOL	3.810405	4	0.4323
WTI VOLUME CHAN	13.29632	4	0.0099
All	33.11505	20	0.0328

Table E4 Stacked Impulse Response Functions — 6-panel chart showing response of each variable to all innovations (Cholesky, 10-day horizon)



F. EViews — Variance Decomposition

Figure F1 Variance Decomposition — Stacked bar chart (6 panels, periods 1–10, all 6 variables)

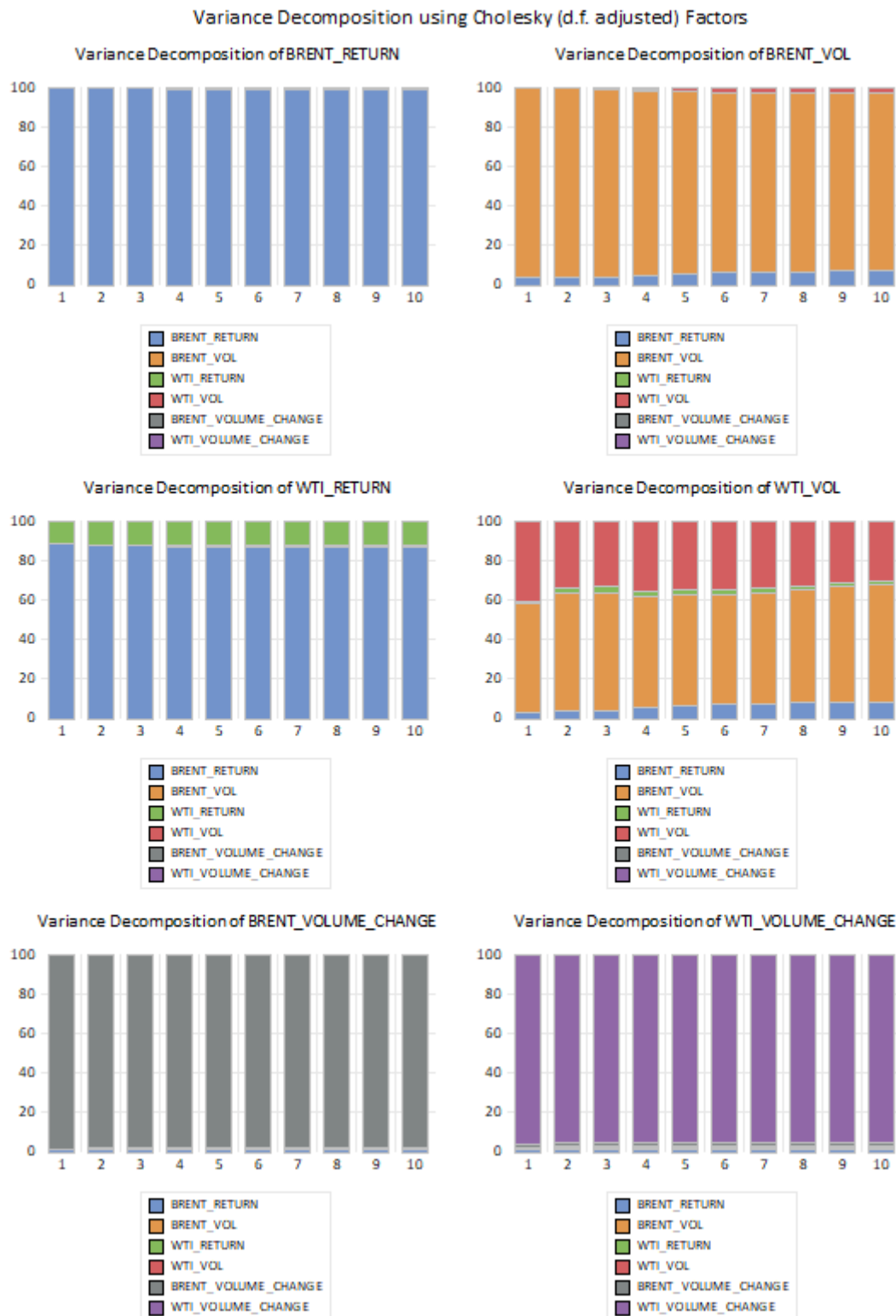


Table F2 Variance Decomposition Table — Numerical FEVD values periods 1–10 (source for Tables 7, 8, 9 and all three TCIs: Returns 43.42%, Volatility 31.56%, Liquidity 1.37%)

Variance Decomposition using Cholesky (d.f. adjusted) Factors

Variance Decomposition of BRENT_RETURN:							
Period	S.E.	BRENT_RE	BRENT_VOL	WTI_RETU	WTI_VOL	BRENT_VO	WTI_VOLU
1	0.023413	100.0000	0.000000	0.000000	0.000000	0.000000	0.000000
2	0.023429	99.86896	0.031939	0.002703	1.36E-05	0.063370	0.033013
3	0.023495	99.32326	0.063785	0.023314	0.154918	0.401863	0.032863
4	0.023547	99.04884	0.266124	0.070115	0.176247	0.401865	0.036806
5	0.023558	98.97953	0.282079	0.092083	0.181716	0.409912	0.054678
6	0.023560	98.96238	0.287039	0.092069	0.182712	0.419600	0.056196
7	0.023563	98.94588	0.287347	0.093482	0.192080	0.419573	0.061639
8	0.023565	98.92885	0.290066	0.095027	0.201110	0.422972	0.061974
9	0.023567	98.91544	0.291636	0.097887	0.209127	0.423157	0.062752
10	0.023569	98.90020	0.293977	0.098268	0.221290	0.423183	0.063079
Variance Decomposition of BRENT_VOL:							
Period	S.E.	BRENT_RE	BRENT_VOL	WTI_RETU	WTI_VOL	BRENT_VO	WTI_VOLU
1	0.027264	2.757308	97.24269	0.000000	0.000000	0.000000	0.000000
2	0.041031	2.705886	96.69069	0.019062	0.445546	0.132965	0.005849
3	0.051588	3.287727	95.61123	0.058002	0.951309	0.085552	0.006185
4	0.060628	4.142298	94.08305	0.052858	1.653162	0.063686	0.004949
5	0.068373	4.761835	92.92689	0.055392	2.196819	0.050413	0.008655
6	0.075148	5.272592	92.01790	0.047248	2.608410	0.043960	0.009895
7	0.081194	5.671509	91.31453	0.041571	2.922671	0.040472	0.009247
8	0.086688	5.998594	90.79572	0.038050	3.122077	0.036810	0.008747
9	0.091687	6.246762	90.45505	0.036735	3.219229	0.033954	0.008270
10	0.096286	6.442851	90.22588	0.036881	3.254575	0.031742	0.008069
Variance Decomposition of WTI_RETURN:							
Period	S.E.	BRENT_RE	BRENT_VOL	WTI_RETU	WTI_VOL	BRENT_VO	WTI_VOLU
1	0.025570	88.10806	0.030583	11.86136	0.000000	0.000000	0.000000
2	0.025688	87.32445	0.046532	12.59589	0.004455	0.005740	0.022942
3	0.025736	87.01686	0.047395	12.55099	0.226852	0.133745	0.024159
4	0.025795	86.85585	0.152275	12.56965	0.243589	0.144385	0.034242
5	0.025810	86.80704	0.185402	12.55696	0.253559	0.160239	0.036791
6	0.025814	86.78613	0.200162	12.55878	0.256265	0.161066	0.037598
7	0.025817	86.77094	0.205381	12.55559	0.267908	0.161408	0.038768
8	0.025818	86.76060	0.214092	12.55566	0.267940	0.162858	0.038855
9	0.025820	86.74974	0.223901	12.55466	0.269726	0.162855	0.039119
10	0.025822	86.73686	0.234229	12.55280	0.273875	0.163086	0.039157
Variance Decomposition of WTI_VOL:							
Period	S.E.	BRENT_RE	BRENT_VOL	WTI_RETU	WTI_VOL	BRENT_VO	WTI_VOLU
1	0.029794	2.058222	56.18210	0.479316	41.28037	0.000000	0.000000
2	0.043432	2.708044	60.62689	2.740998	33.89174	0.022088	0.010239
3	0.055081	3.395078	59.70522	3.355858	33.51825	0.014236	0.011367
4	0.066038	4.406619	56.80821	2.807119	35.95099	0.011894	0.015167
5	0.074870	5.457241	56.52604	2.630104	35.36454	0.010214	0.011858
6	0.082821	6.148872	56.41555	2.435359	34.97874	0.011761	0.009720
7	0.090070	6.719427	56.69842	2.233149	34.32891	0.011583	0.008510
8	0.096504	7.182597	57.59996	2.093839	33.10447	0.011165	0.007968
9	0.102426	7.513187	58.65762	1.936991	31.87350	0.011283	0.007420
10	0.107906	7.780975	59.86193	1.793275	30.54583	0.011271	0.006718
Variance Decomposition of BRENT_VOLUME_CHANGE:							
Period	S.E.	BRENT_RE	BRENT_VOL	WTI_RETU	WTI_VOL	BRENT_VO	WTI_VOLU
1	0.510333	0.193522	0.338698	0.030572	0.180996	99.25621	0.000000
2	0.560523	0.200326	0.487846	0.046239	0.174472	99.08126	0.009856
3	0.563003	0.239218	0.564390	0.089801	0.241314	98.85243	0.012851
4	0.564848	0.337043	0.586360	0.113878	0.254074	98.52819	0.180454
5	0.565252	0.337924	0.590490	0.188089	0.264940	98.42706	0.191502

Variance Decomposition using Cholesky (d.f. adjusted) Factors

6	0.567806	0.335407	0.606753	0.187005	0.265011	98.20637	0.399457
7	0.567928	0.335264	0.607193	0.189139	0.265962	98.19519	0.407253
8	0.568031	0.335173	0.612259	0.190149	0.266050	98.17748	0.418890
9	0.568125	0.335559	0.615158	0.190128	0.266961	98.15659	0.435599
10	0.568133	0.335735	0.615371	0.190192	0.267168	98.15420	0.437330
Variance Decomposition of WTI VOLUME CHANGE:							
Period	S.E.	BRENT_RE	BRENT_VOL	WTI_RETU	WTI_VOL	BRENT_VO	WTI_VOLU
1	0.407976	0.174434	0.955199	0.125389	0.003495	2.197087	96.54440
2	0.418057	0.247071	1.340257	0.233361	0.037755	2.464373	95.67718
3	0.440086	0.239328	1.281642	0.288648	0.053968	2.289541	95.84687
4	0.440704	0.255434	1.310866	0.287985	0.063410	2.286434	95.79587
5	0.441009	0.257100	1.315356	0.291884	0.068433	2.305489	95.76174
6	0.443126	0.254665	1.321080	0.290227	0.067831	2.292925	95.77327
7	0.443253	0.254571	1.321065	0.290066	0.071709	2.294349	95.76824
8	0.443638	0.254244	1.324962	0.289825	0.071637	2.292557	95.76678
9	0.443769	0.254359	1.326791	0.289655	0.071613	2.291498	95.76608
10	0.443775	0.254676	1.326944	0.289682	0.072388	2.292214	95.76410
Cholesky One S.D. (d.f. adjusted)							
Cholesky ordering: BRENT RETURN BRENT VOL WTI RETURN WTI VOL BRENT VOLUME CHAN							
GE WTI_VOLUME_CHANGE							

Source Reference — Granger Causality Definition

The block exogeneity Granger causality test used in this study is defined in the official EViews documentation as follows:

“Carries out pairwise Granger causality tests and tests whether an endogenous variable can be treated as exogenous. For each equation in the VAR, the output displays χ^2 (Wald) statistics for the joint significance of each of the other lagged endogenous variables in that equation. The statistic in the last row (All) is the χ^2 -statistic for joint significance of all other lagged endogenous variables in the equation.”

Source: IHS Global Inc. (2020). *EViews 11 User's Guide II, Chapter 44 — Vector Autoregression and Error Correction Models, Section: Pairwise Granger Causality Tests, Page 848.*